

Lesson-End Project

Implementing Web App

Project agenda: To implement web app for hosting a website for your business

Description: Edu-sim an educational technology company, is evaluating Azure web apps to migrate its PHP websites from on-premises Windows servers. The IT team assesses the current setup, checks Azure compatibility, and plans the migration using Azure Migrate. They set up and secure Azure web apps, establish CI/CD pipelines with Azure DevOps, configure deployment slots, and integrate automated testing. After functional, performance, and user acceptance testing, the team deploys to production and implements continuous monitoring.

Tools required: Azure account with administrator access

Prerequisites: Resource group

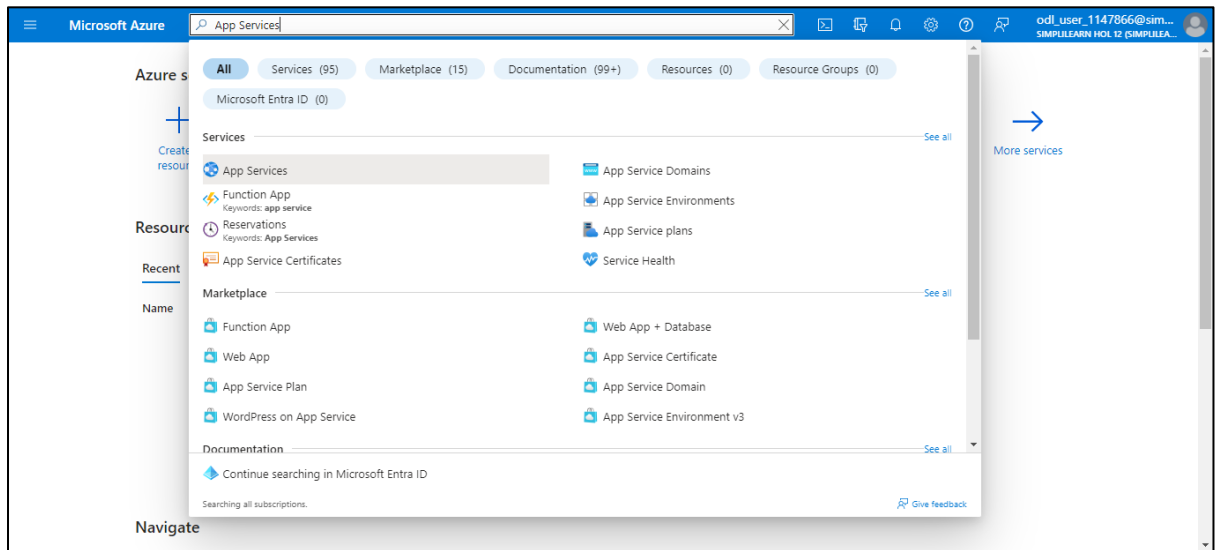
Expected deliverables: Add screenshots of configuring and deploying the web apps and demonstrate the deployment of code to the staging deployment slot

Steps to be followed:

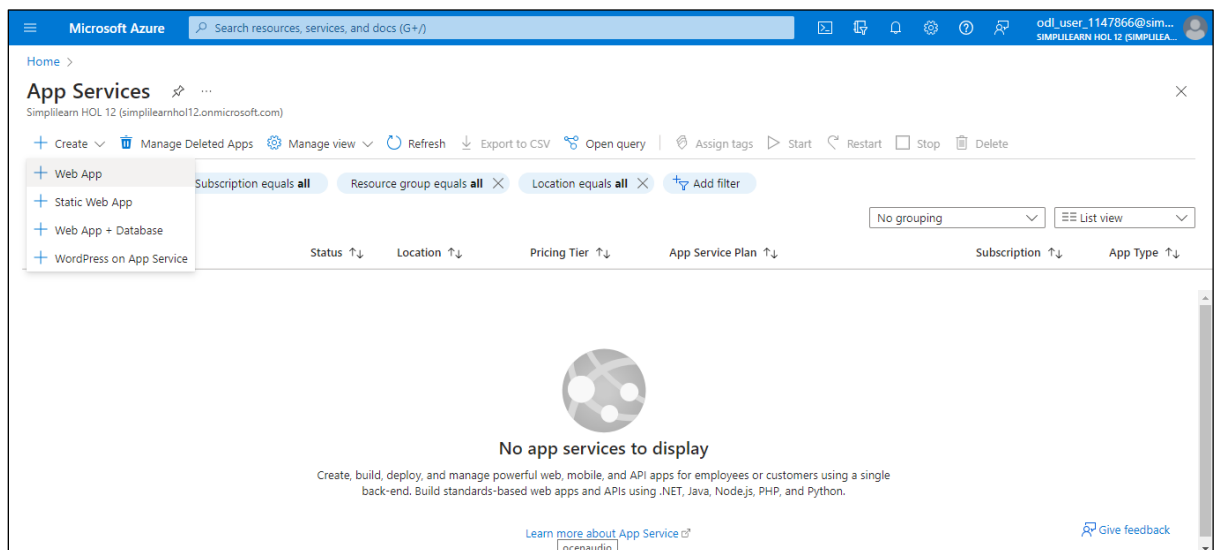
1. Create an Azure web app
2. Create a staging deployment slot
3. Configure web app deployment settings
4. Deploy code to the staging deployment slot

Step 1: Create an Azure web app

1.1 Log in to the **Azure portal**; search for and select **App Services**



1.2 In the **App Services** page, click on **+Create** and select **Web App**



1.3 In the **Create Web App** page, under **Project Details**, specify the information as shown in the screenshot below and click on **Review + create**

The screenshot shows the 'Create Web App' page in the Microsoft Azure portal. The 'Project Details' section is active, showing the following configuration:

- Subscription:** Simplilearn HOL 12
- Resource Group:** (New) az305-rg2
- Instance Details:**
 - Name:** mywbap3
 - Publish:** Code (selected), Docker Container, Static Web App
 - Runtime stack:** PHP 8.2
 - Operating System:** Linux (selected), Windows
 - Region:** East US

At the bottom, there are buttons for 'Review + create', '< Previous', and 'Next: Database >'. A note at the bottom states: 'Not finding your App Service Plan? Try a different region or select your App Service Plan.'

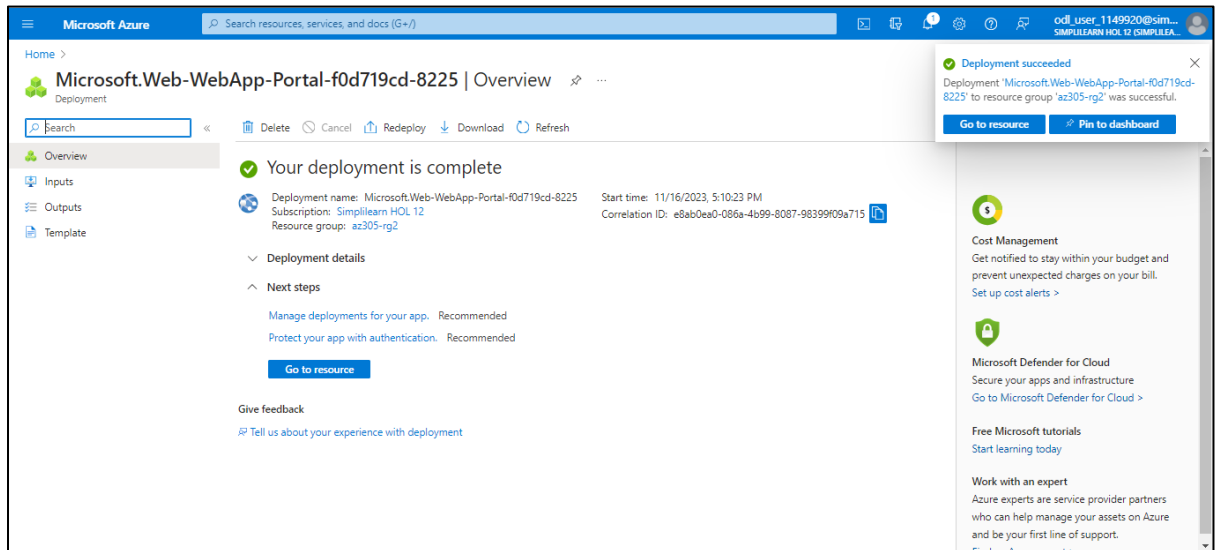
1.4 In the **Review + create** tab, click on **Create**

The screenshot shows the 'Create Web App' page in the Microsoft Azure portal, with the 'Review + create' tab selected. The page displays a summary of the configuration:

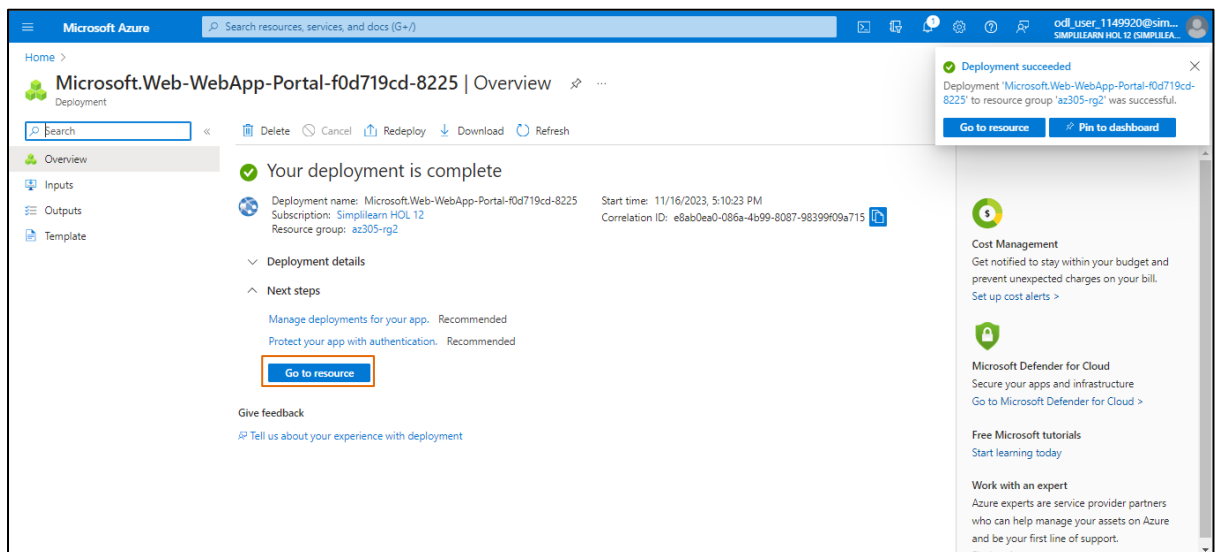
- Summary:** Web App by Microsoft
- Details:**
 - Subscription:** 3b2a36a5-7d37-4026-9bb8-a24abe7ab64e
 - Resource Group:** az305-rg2
 - Name:** mywbap3
 - Publish:** Code
 - Runtime stack:** PHP 8.2
- App Service Plan:**
 - Name:** ASP-az305rg1-830b
 - Operating System:** Linux
 - Region:** East US
 - SKU:** Premium V3
 - Size:** Small

At the bottom, there are buttons for 'Create', '< Previous', and 'Next >'. A link 'Download a template for automation' is also present.

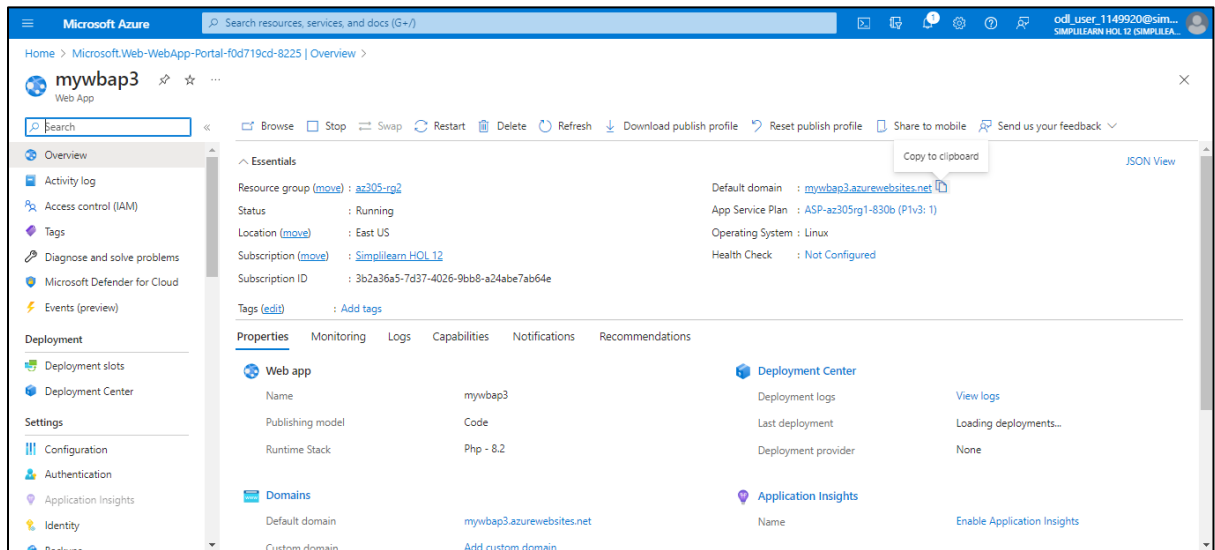
The deployment completion screen appears.



1.5 In the **Overview** page, click on **Go to resource**

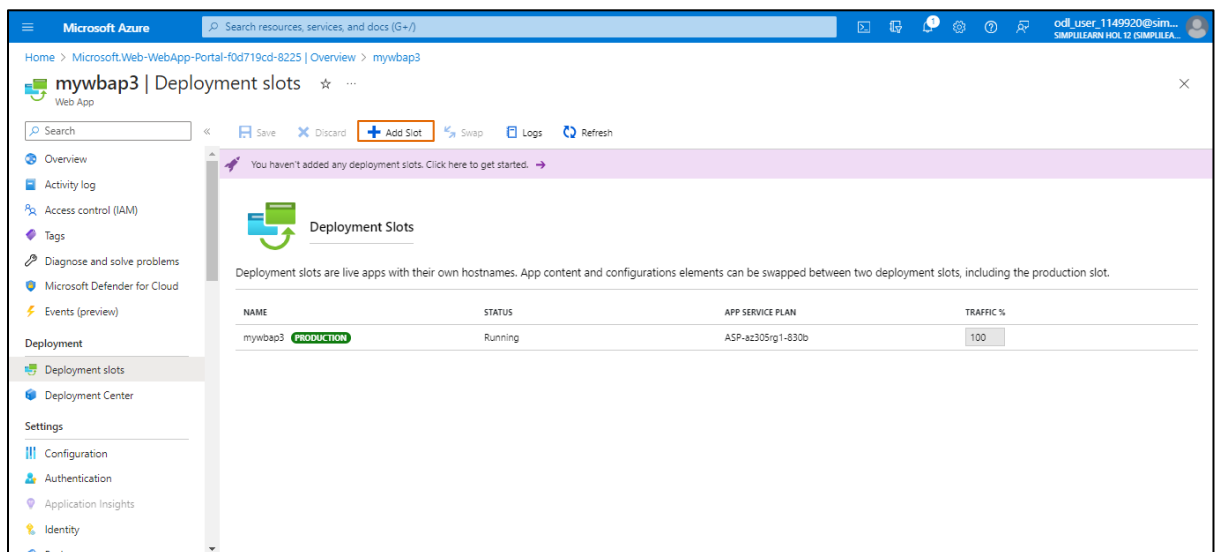


In the newly developed **Web App** page (**mywbap3**), the **Default domain** URL appears:

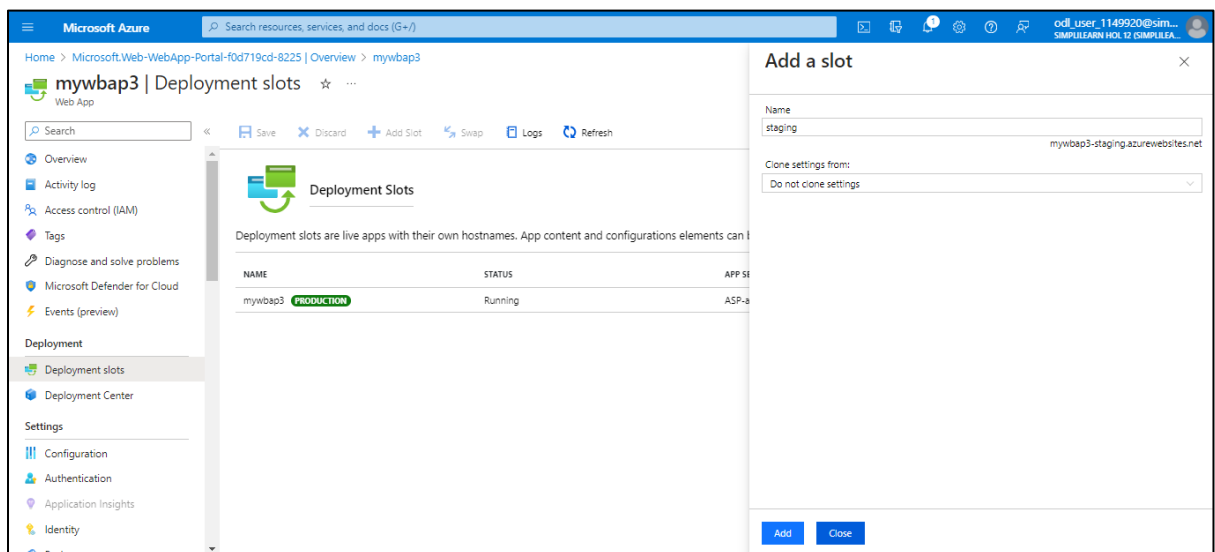


Step 2: Create a staging deployment slot

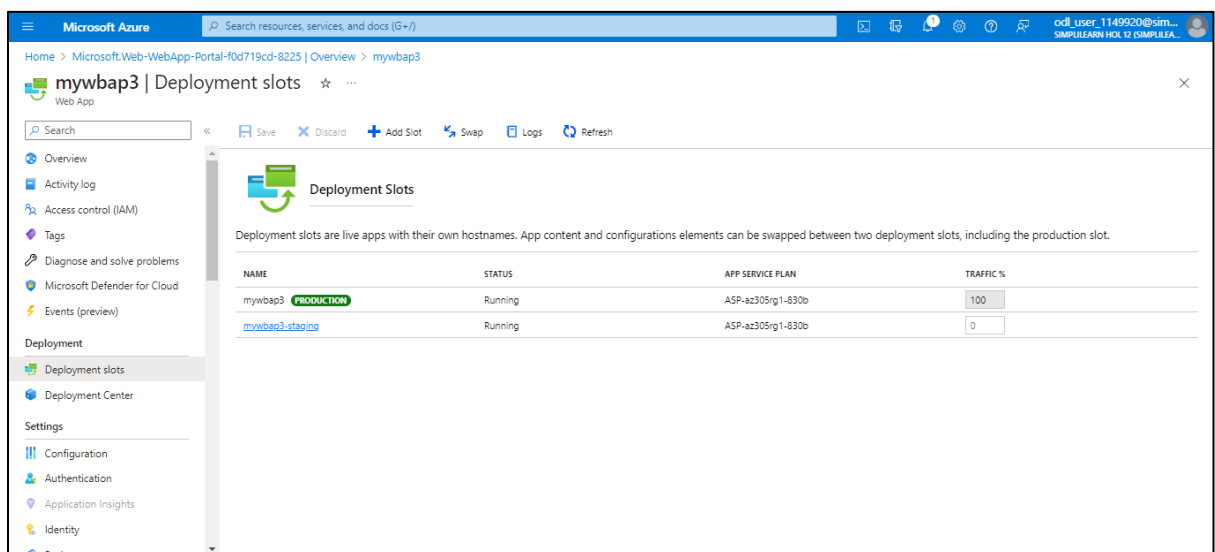
2.1 Under the **Deployment** section of the left side pane, click on **Deployment slots**. Then, select **Add Slot**.



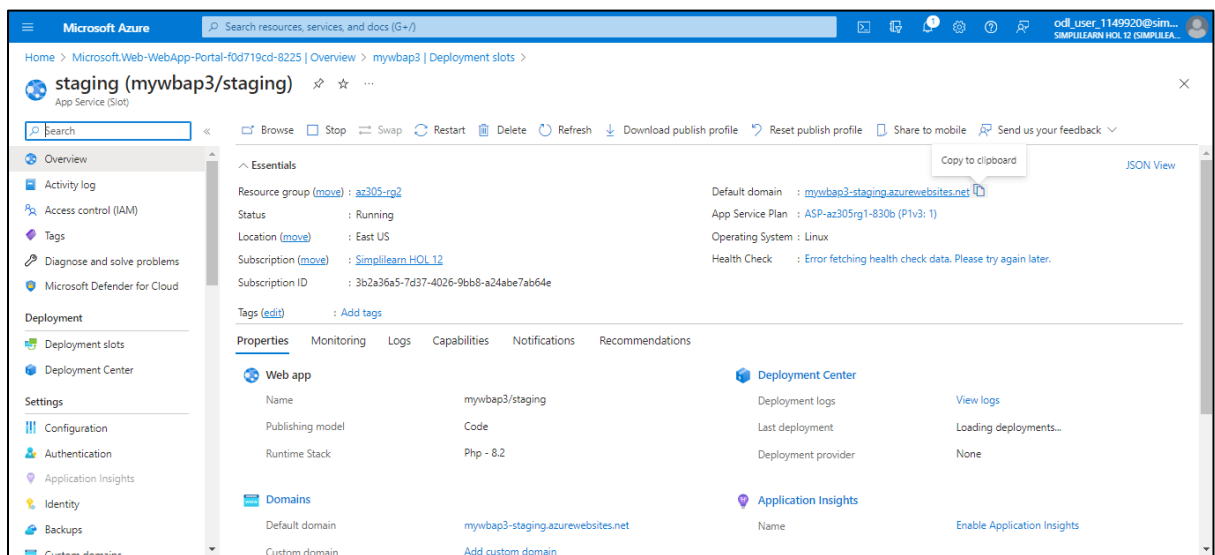
2.2 In the **Add a slot** page, enter the **Name** as **staging** and choose the **Clone settings** from as **Do not clone settings**. Once the details are entered, click **Add** and select **Close**.



2.3 Navigate back to the **Deployment slots** page and click on the entry representing the newly created staging slot (**mywbap3-staging**)

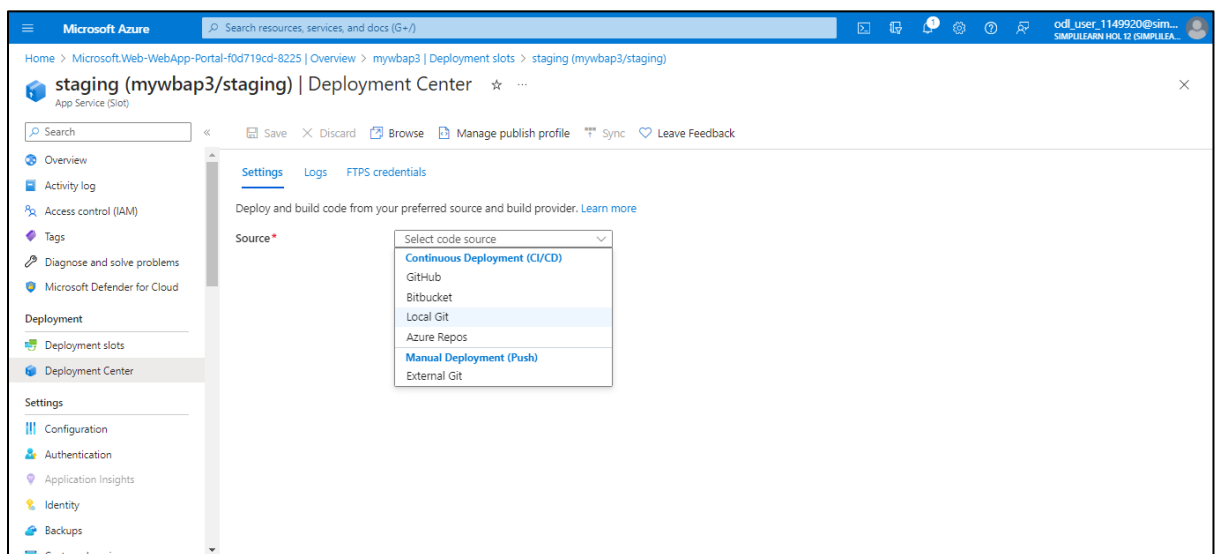


Review the essentials in the **App Service (Slot)** page and note that the **Default domain** URL differs from the one assigned to the production slot.

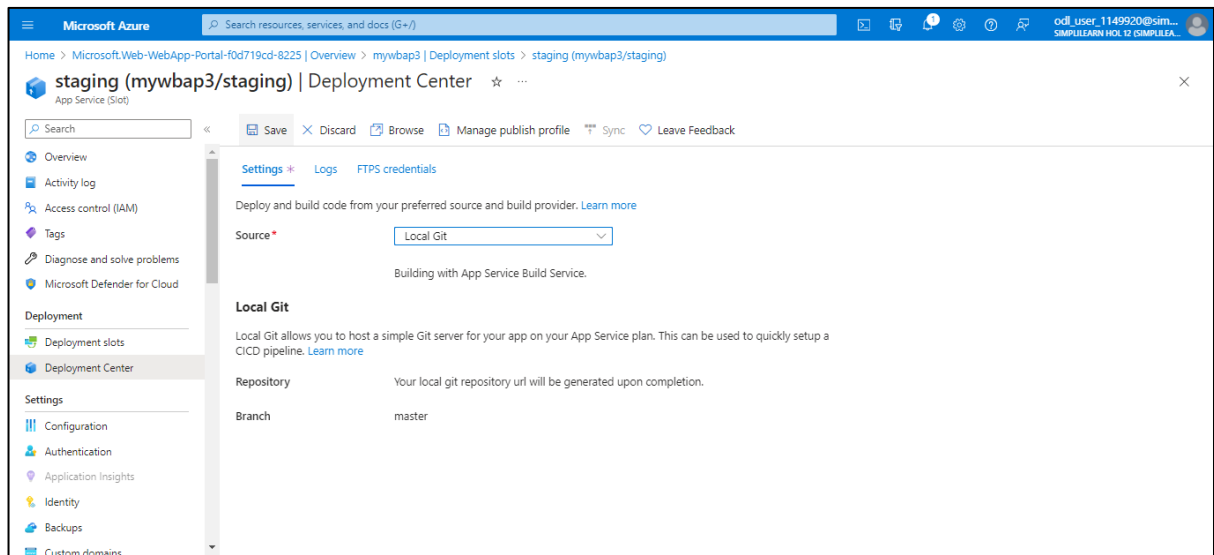


Step 3: Configure web app deployment settings

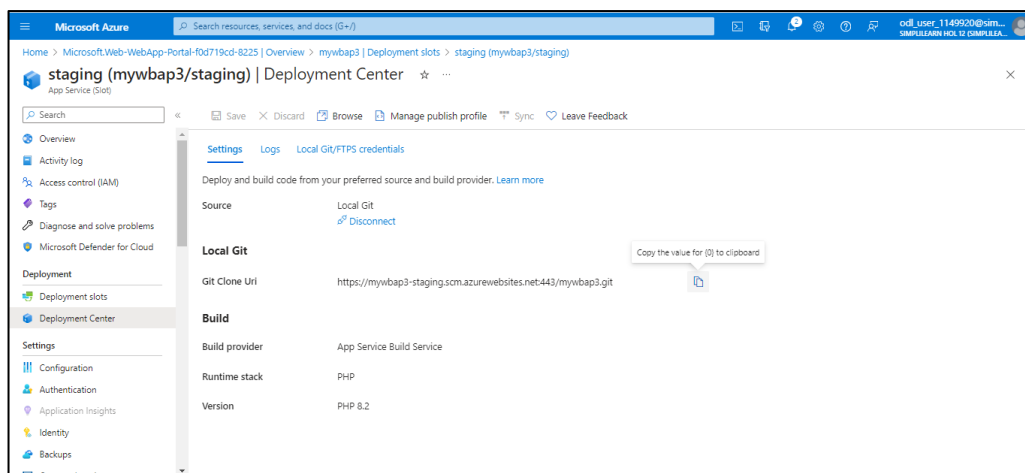
3.1 Under the **Deployment** section, click on **Deployment Center** and then select the **Settings** tab. In the **Source** drop-down list, select **Local Git**.



3.2 Click on **Save**

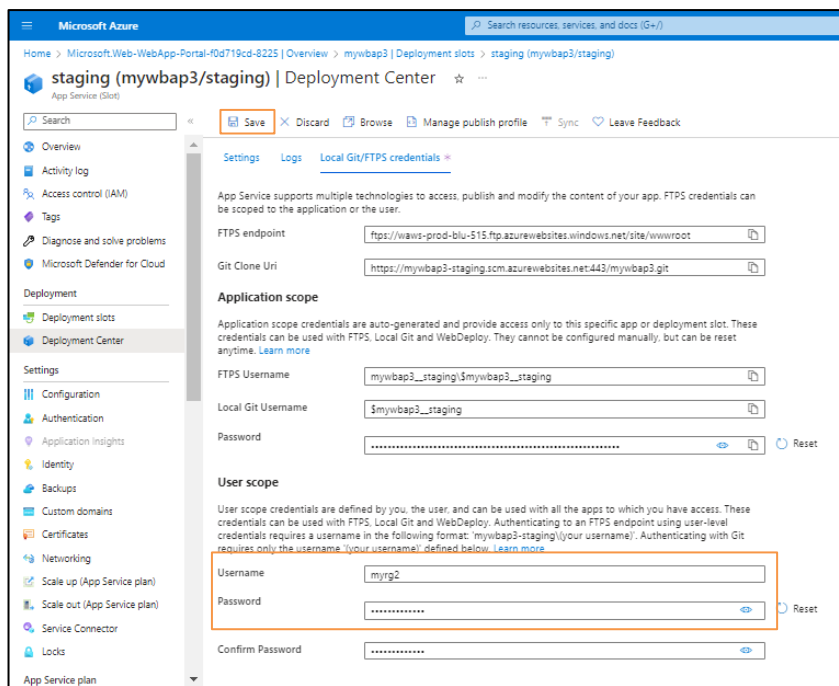


3.3 Under the **Local Git**, copy the **Git Clone Url** to your Notepad



Note: You will need the **Git Clone Url** value in the next task of this lab.

3.4 In the **Deployment Center** page, select the **Local Git/FTPS credentials** tab. Under the **User scope** section, specify the **Username** and **Password** and click on **Save**.



The screenshot shows the Microsoft Azure portal interface for the 'staging (mywbap3/staging)' deployment slot. The 'Local Git/FTPS credentials' tab is selected. The 'User scope' section is highlighted with an orange box, indicating where to enter the username and password for deployment.

Deployment Center | **Local Git/FTPS credentials**

App Service supports multiple technologies to access, publish and modify the content of your app. FTPS credentials can be scoped to the application or the user.

FTPS endpoint:

Git Clone Uri:

Application scope

Application scope credentials are auto-generated and provide access only to this specific app or deployment slot. These credentials can be used with FTPS, Local Git and WebDeploy. They cannot be configured manually, but can be reset anytime. [Learn more](#)

FTPS Username:

Local Git Username:

Password:

User scope

User scope credentials are defined by you, the user, and can be used with all the apps to which you have access. These credentials can be used with FTPS, Local Git and WebDeploy. Authenticating to an FTPS endpoint using user-level credentials requires a username in the following format: 'mywbap3-staging/your username'. Authenticating with Git requires only the username, 'your username', defined below. [Learn more](#)

Username:

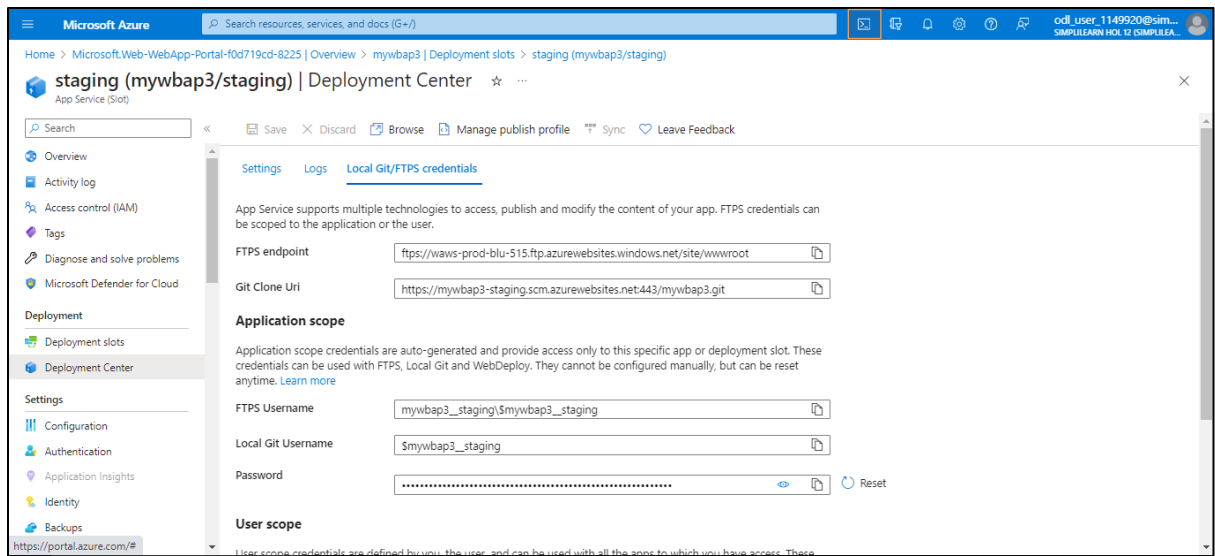
Password:

Confirm Password:

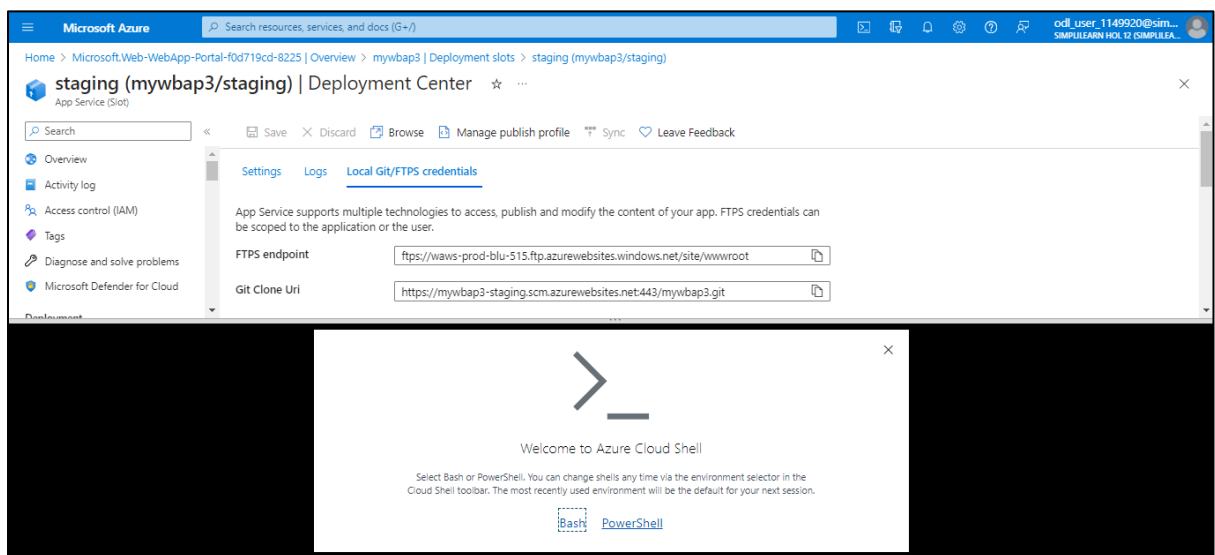
Note: Copy these credentials. Users will need it for the next task in this lab.

Step 4: Deploy code to the staging deployment slot

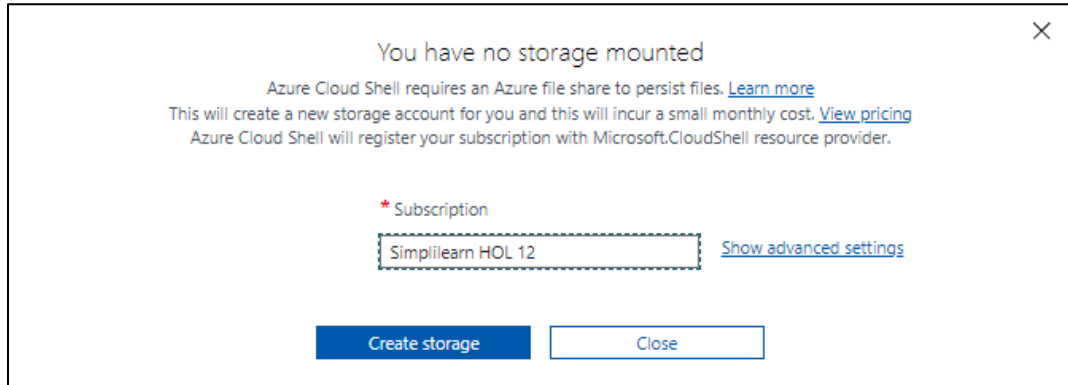
4.1 Click on the **Cloud Shell** icon near the search bar



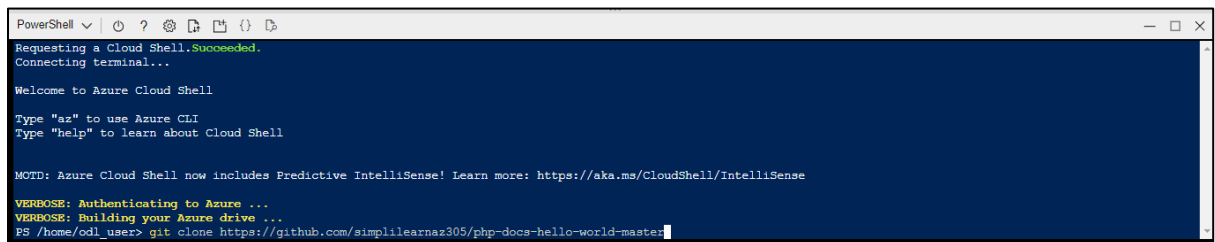
4.2 If prompted to select either **Bash** or **PowerShell**, select **PowerShell**



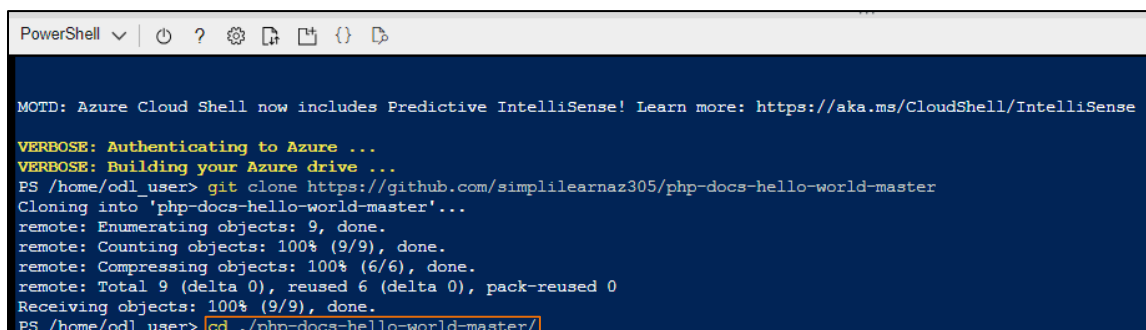
- 4.3 For first time users, **Cloud Shell** are presented with the **You have no storage mounted** message. Select the subscription you are using in this lab and click on **Create storage**.



- 4.4 From the **PowerShell** pane, run the following command to clone the remote repository containing the code for the web app:
git clone <https://github.com/simplilearnaz305/php-docs-hello-world-master>



- 4.5 Run the following command to set the current location to the newly created clone of the local repository:
cd ./php-docs-hello-world-master/



4.6 Run the following command to add the remote git:

```
git remote add [deployment_user_name] [git_clone_url]
```

```
PowerShell v | ? ? ? ? ? ? ? ?
```

MOTD: Azure Cloud Shell now includes Predictive IntelliSense! Learn more: <https://aka.ms/CloudShell/IntelliSense>

VERBOSE: Authenticating to Azure ...
VERBOSE: Building your Azure drive ...
PS /home/odl_user> git clone https://github.com/simplilearnaz305/php-docs-hello-world-master
Cloning into 'php-docs-hello-world-master'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 9 (delta 0), reused 6 (delta 0), pack-reused 0
Receiving objects: 100% (9/9), done.
PS /home/odl_user> cd ./php-docs-hello-world-master/
PS /home/odl_user/php-docs-hello-world-master> git remote add myrg2 https://mywabp3-staging.scm.azurewebsites.net:443/mywabp3.git

Note: Make sure to replace the [deployment_user_name] and [git_clone_url] placeholders with the value of the **Deployment Credentials** username and **Git Clone Url** respectively, which you identified in the previous task.

4.7 Run the following commands one by one to push the sample web app code from the local repository to the Azure web app staging deployment slot:

```
git checkout -b master
```

```
git push -u [deployment_user_name] master
```

Type the **[deployment_user_name]** and the corresponding **password** (Set in the previous task)

```
error: remote myrg2 already exists.
PS /home/odl_user/php-docs-hello-world-master> git checkout -b master
Switched to a new branch 'master'
PS /home/odl_user/php-docs-hello-world-master> git push -u myrg2 master
Username for 'https://mywbap3-staging.scm.azurewebsites.net:443': myrg2
Password for 'https://myrg2@mywbap3-staging.scm.azurewebsites.net:443':
^[D^[[C^[[DEnumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 3 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (9/9), 2.09 KiB | 2.09 MiB/s, done.
Total 9 (delta 0), reused 9 (delta 0), pack-reused 0
remote: Deploy Async
remote: Updating branch 'master'.
```

Notes:

1. The value following git remote add does not have to match the **Deployment Credentials** username but must be unique.
2. Make sure to replace the [deployment_user_name] placeholder with the value of the **Deployment Credentials** username, which you identified in the previous task.

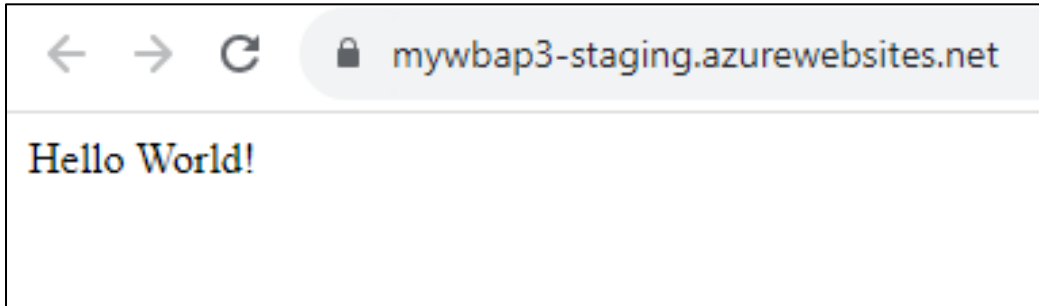
The code is deployed as shown in the screenshot below:

```
remote: Found 0 issue(s)
remote:
remote: Build Summary :
remote:
remote: Errors (0)
remote: Warnings (0)
remote:
remote: Triggering recycle (preview mode disabled).
remote: Deployment successful. deployer = deploymentPath =
remote: Deployment Logs : 'https://mywebap3-staging.scm.azurewebsites.net/newui/jsonviewer?view_url=/api/deployments/11870df51c324127cb19c9747d57c8899f9b32a5/log'
To https://mywebap3-staging.scm.azurewebsites.net:443/mywebap3.git
 * [new branch]      master -> master
Branch 'master' set up to track remote branch 'master' from 'myrg2'.
PS /home/odl_user/php-docs-hello-world-master>
```

4.8 Close the **Cloud Shell** pane. On the **staging** page, select **Overview** and then click on the **Default domain** URL to display the default web page in a new browser tab.

The screenshot shows the Microsoft Azure portal interface for a Web App named 'staging (mywebap3/staging)'. The left sidebar contains navigation options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Deployment, Deployment slots, Deployment Center, Settings, Configuration, Authentication, Application Insights, Identity, and Backups. The main content area displays the 'Overview' tab with 'Essentials' and 'Properties' sections. The 'Essentials' section includes details like Resource group (az305-rg2), Status (Running), Location (East US), Subscription (Simplilearn HOL 12), and Subscription ID (3b2a36a5-7d37-4026-9bb8-a24abe7ab64e). The 'Properties' section shows the Web app name (mywebap3/staging), Publishing model (Code), and Runtime Stack (Php - 8.2). The 'Domains' section shows the Default domain (mywebap3-staging.azurewebsites.net). The 'Deployment Center' section shows the Deployment logs and Last deployment status.

Verify that the browser page displays **Hello World!** message and close the new tab



By following these steps, you have successfully created an Azure web app and a staging deployment slot. You have also configured web app deployment settings and deployed code to the staging deployment slot.